

Structure, Dynamics and Vibrational Spectroscopy of Interfacial Aqueous Solutions

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Main arguments:

- (1) **Ab initio molecular dynamics simulations** → **The experimental spectra**;
- (2) **Hydrogen Bond (HB) dynamics + Instantaneous interface** → **Thickness of interfaces**;
- (3) **HB population + RNN** → **Classification of dynamical processes**.

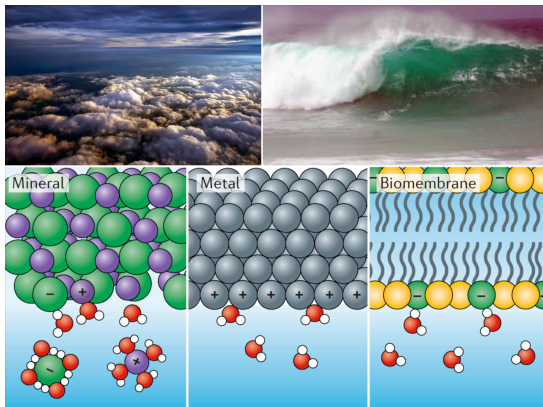
► Introduction

► **Methods:** Vibrational Sum-Frequency Generation (VSFG or SFG); Density Functional Theory-based Molecular Dynamics (DFTMD), Vibrational Density of States, HB dynamics

► **Results:** VSFG of alkali nitrate and alkali iodide solutions, Instantaneous interfaces and thickness of interfaces, Structural and features of solution/vapor interface; Classification of dynamical processes.

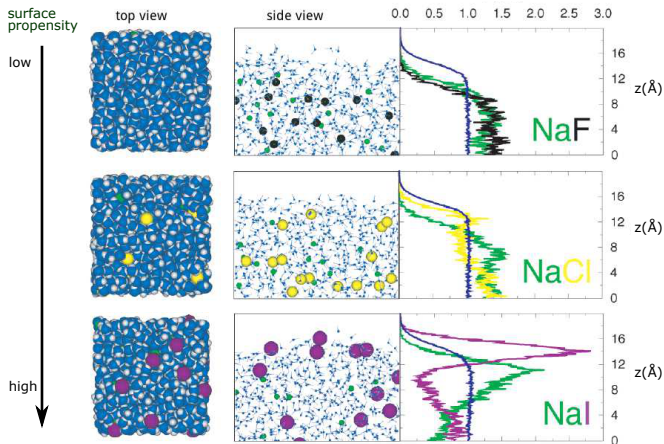
► Conclusions and perspectives

Aqueous interfaces are ubiquitous.



Aqueous-vapor interfaces: terminated HB network, statistical distribution of the position and orientation of molecules (Gonella, et al, *Nature Rev. Chem.* **5**, 466 (2021)).

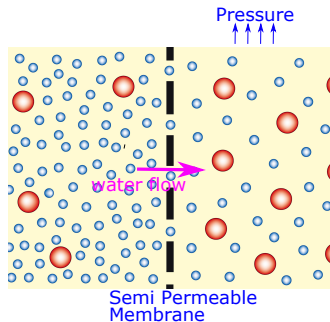
Ions have different surface propensity.



Ions' distribution with depth at the water-vapor interface are different. (Jungwirth, et al *Phys. Chem.* **105**, 10468 (2001)). Ions affect properties of interfaces significantly.

Applications and problems

- Microscopic understanding of interfaces of aqueous solutions is essential to further energy and human development.

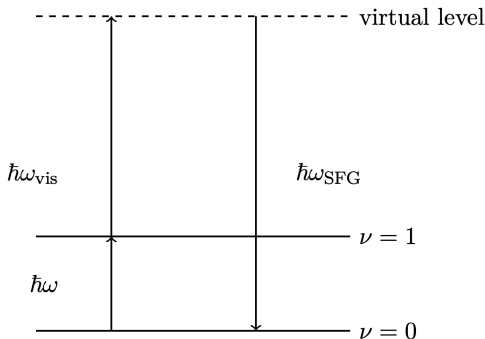
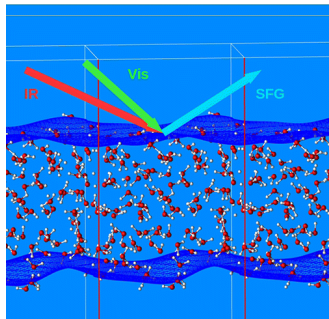


The water flow in Osmotic power plant.

- When the solution contains **larger molecular anions** such as **nitrate, sulfate and carbonate ions**, the processes occurring at the interface need to be further studied. (Gopalakrishnan et al, *J. Phys. B* **109** (2005); Jungwirth and Tobias, *Chem. Rev.*, **106**, 1259 (2006))

Selectively probing the interface: VSFG spectroscopy

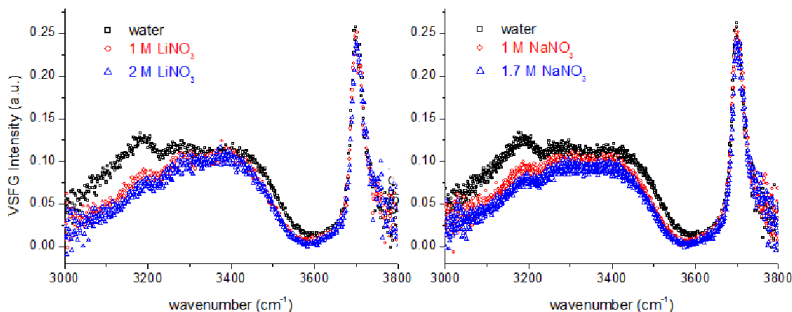
- ▶ VSFG (SFG) is used to get **vibrational information** of water molecules at the water/vapor interface (Shen, et al, *Chem. Phys. Lett.*, **133**, 189 (1987)).
- ▶ Sensitive only to regions of a material where **inversion symmetry is broken**.



VSFG process which involves IR and Raman transitions (Sulpizi, Salanne, Sprik and Gaigeot, *J. Phys. Chem. Lett.*, **83** (2013)).

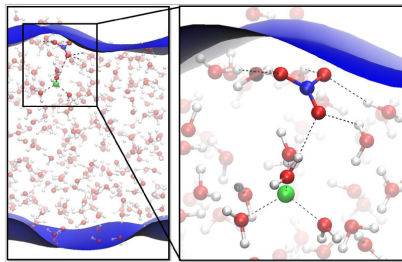
VSFG spectra: Ions modify hydrogen bond structure at the interfaces.

The intensity spectra of nitrate solution show a **depletion of the 3200 cm^{-1} region** in the experiments (Hua, Verreault and Allen, *J. Phys. Chem. C*, **118**, 24941 (2014)).



ab initio Molecular Dynamics (AIMD)

- ▶ Molecular Dynamics: **Born-Oppenheimer dynamics**;
- ▶ Exchange-Correlation functional: **BLYP**;
- ▶ Basis sets: **DZVP-MOLOPT-GTH**;
- ▶ Plane wave cut off: **280 Ry**;
- ▶ NVT ensemble ($T = 300$ K);
- ▶ **Grimme correction**;
- ▶ PBC Box: $19.7 \times 19.7 \times 40.0$ ($\text{\AA}^3$); Time step: 0.5 fs; Simulation time: **60 ps**.



The lithium and nitrate ions are **separated by water** and remain in this configuration stably at the interface of LiNO_3 solution.

VSFG spectra from molecular dynamics simulations

- **Intensity** of the SFG spectra is proportional to 2nd-order susceptibility's absolute value squared, $|\chi^{(2)}(\omega)|^2$:

$$\chi_{ssp}^{(2)}(\omega) = \frac{i\omega}{k_B T} \sum_{j=1}^{N_m} \int dt \langle A_{xx}^j(t) M_z^j(0) \rangle e^{i\omega t}. \quad (1)$$

(Morita, et al, *Phys. Chem. Chem. Phys.*, **10**, 5801 (2008))

- Assume

(1) Bond elongation is **small**;

(2) **Derivatives** of dipole moments/polarizability w.r.t. bond length **vary slowly**:

$$\begin{aligned} \chi_{ssp}^{(2)}(\omega) = & \frac{i\omega}{k_B T} \frac{1}{\pm\omega^2} \sum_{j=1}^{N_m} \sum_{a=1}^{N_{modej}} \left\langle \frac{\partial A_{xx}^j(0)}{\partial q_a} \right\rangle \left\langle \frac{\partial M_z^j(0)}{\partial q_a} \right\rangle \\ & \times \int e^{i\omega t} \left\langle \frac{dq_a(t)}{dt} \frac{dq_a(0)}{dt} \right\rangle. \end{aligned} \quad (2)$$

(Khatib and Sulpizi, *J. Phys. Chem. Lett.*, **8** (6), 1310 (2017))

Vibrational density of states (VDOS)

The calculations and analysis of the Vibrational Density Of States (VDOS) permits to obtain **information on the HB strength**.

Correlation functions of atomic velocity in a system of N atoms:

$$C_{(z)}(t) = \frac{\langle \sum_{i=1}^N \vec{v}_{i,(z)}(t) \cdot \vec{v}_{i,(z)}(0) \rangle}{\langle \sum_{i=1}^N \vec{v}_i(0) \cdot \vec{v}_i(0) \rangle},$$

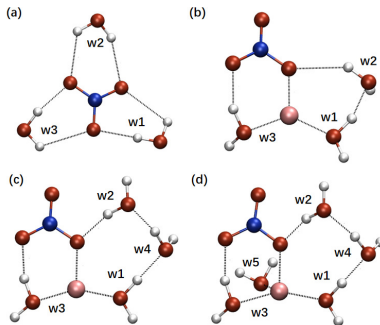
VDOS $g(\nu)$ ($g_z(\nu)$):

$$g_{(z)}(\nu) = \sqrt{\frac{2}{\pi}} \int_0^\infty dt \cos(2\pi\nu t) C_{(z)}(t).$$

1. VDOS and SFG spectra

Cluster models sufficient to explain the spectral features?

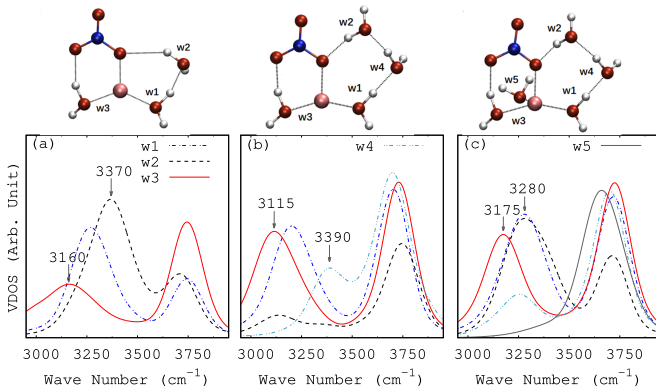
Geometry optimized clusters of water molecules with nitrate and alkali metal ions and increasing number of waters.



1. VDOS and SFG spectra

Increasing the number of waters: $\text{LiNO}_3(\text{H}_2\text{O})_n$ ($n = 3 - 5$), strong red-shift

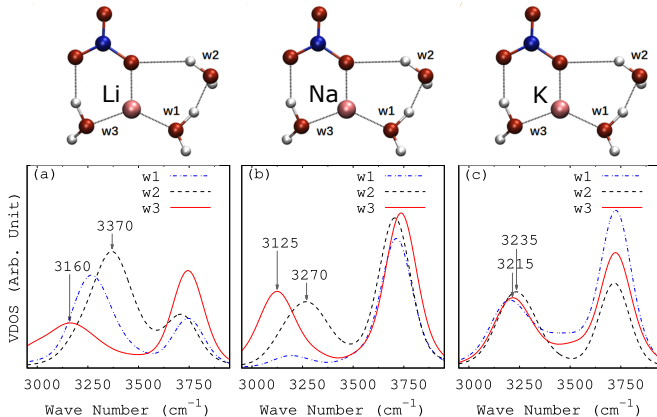
The OH stretching peaks in the HB region is quite strong **red-shifted** around 3200 cm^{-1} . This is **not observed in the experiment!**



1. VDOS and SFG spectra

Specific cation effect: $\text{RNO}_3(\text{H}_2\text{O})_3$ ($\text{R}=\text{Li}, \text{Na}, \text{K}$), also quite red-shifted

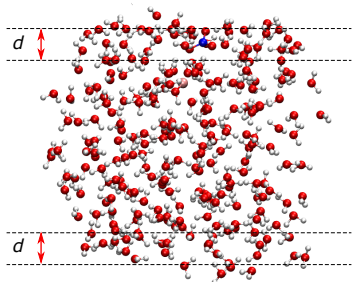
The OH stretching peaks in the HB region are **also quite red-shifted**. Not observed in the experiment!



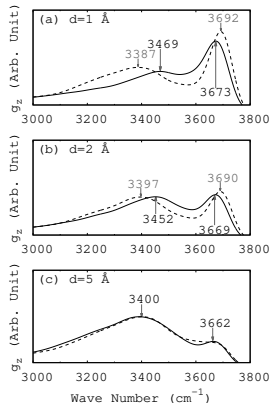
1. VDOS and SFG spectra

A more complete model: the aqueous/vapor interface

The salty interface has a blue shifted HB band. → Aqueous/vapor interface is a better model to calculate the SFG spectra.



Definition of horizontal interface and the thickness for LiNO_3 /vapor interface system by Gibbs dividing surface.

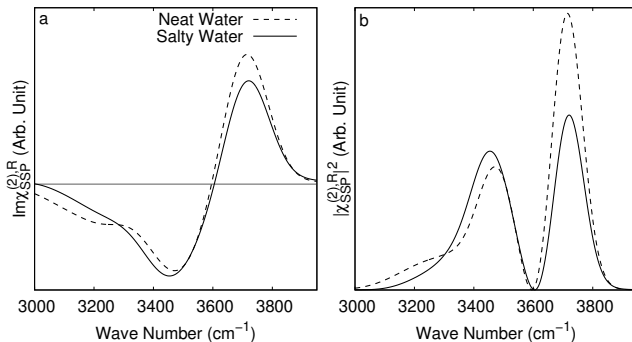


VDOS for water at LiNO_3 interface (solid line) and water/vapor interface (dashed line).

1. VDOS and SFG spectra

Calculated SFG spectra: Consistent with experiment!

The intensity spectra show a **depletion** of the 3200 cm^{-1} region as in the experiments.



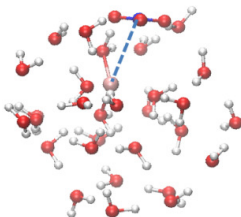
Calculated SFG intensity for LiNO_3 solution shows: 1. **Lower intensity** around 3200 cm^{-1} comes from nitrate at the interface; 2. **Nitrate** prefer to **stay at surface**. (G. Huang, *PhD Dissertation*, Johannes Gutenberg-Universität Mainz, <http://doi.org/10.25358/openscience-6865>)

1. VDOS and SFG spectra

Connecting the cluster and interface view (1)

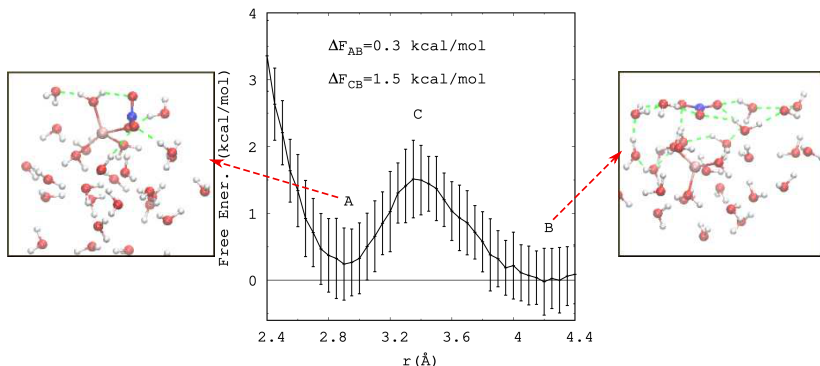
Question: Is it possible to find a contact ion pair at the interface?

- ▶ Use cluster $\text{LiNO}_3(\text{H}_2\text{O})_{30}$ to model the interface with constraint MD.
- ▶ **Reaction coordinate** (r): distance $d_{\text{Li-N}}$.
- ▶ **Calculate free energy difference** between the **contact ion pair** and **separated ion pair** at the interface.



1. VDOS and SFG spectra

Connecting the cluster and interface view (2)



Free energy as a function of the reaction coordinate $d_{\text{Li-N}}$.

- ▶ Two minimums exist for configs A and B.
- ▶ **Separated ion pair** (config. B) is more stable than **contact ion pair** (config. A).

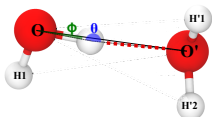
(G. Huang, *PhD Dissertation, Johannes Gutenberg-Universität Mainz*,
<http://doi.org/10.25358/openscience-6865>)

3. HB dynamics of the interface of electrolyte solutions

Definitions of hydrogen bond (HB) and HB population

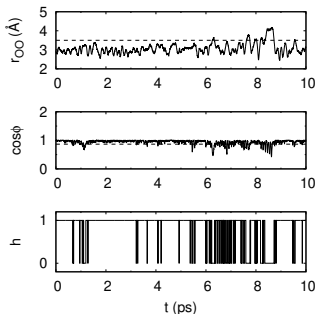
For a particular **tagged pair** of molecules i, j , define

$$h[r(t)] = \begin{cases} 1 & i, j \text{ are H-bonded} \\ 0 & \text{otherwise} \end{cases}$$



Hydrogen bonds. **ADH** criterion:
 $|OO'| < 3.5 \text{ \AA}$ and the $\text{H2-O} \cdots \text{O}'$
 angle $\phi < 30^\circ$; **AHD** criterion:
 $|OO'| < 3.5 \text{ \AA}$ and the $\text{O-H2} \cdots \text{O}'$
 angle $\theta > 120^\circ$.

(Luzar, Chandler, *Nature*, **379**, 55
 (1996))



The **HB population** $h(t)$.

3. HB dynamics of the interface of electrolyte solutions

Correlation functions: $c(t)$ and $s(t)$

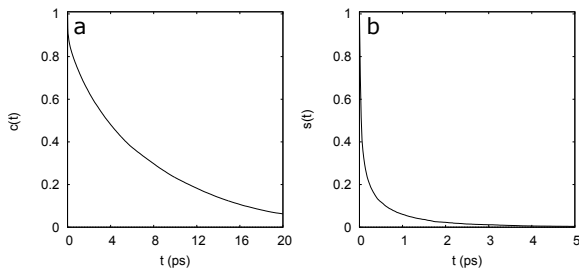
$$c(t) = \langle h(0)h(t) \rangle / \langle h \rangle,$$

$$s(t) = \langle h(0)H(t) \rangle / \langle h \rangle.$$

$\langle \dots \rangle$: an average over all molecular pairs,

$$H(t) = \begin{cases} 1 & i, j \text{ are continuously H-bonded in time interval } [0, t], \\ 0 & \text{otherwise.} \end{cases}$$

(Luzar, *J. Chem. Phys.* **113**, 10663 (2000))



Calculated $c(t)$ and $s(t)$ for bulk water.

3. HB dynamics of the interface of electrolyte solutions

Correlation functions: $k(t)$

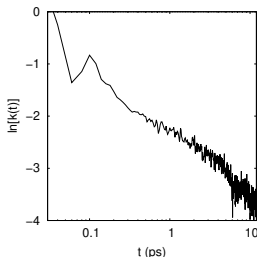
- Reactive flux $k(t)$.

$$k(t) = -\frac{dc(t)}{dt}; n(t) = \langle h(0)[1 - h(t)]h^{(d)}(t) \rangle / \langle h \rangle;$$

$$k(t) = kc(t) - k'n(t) \xrightarrow[\text{fit}]{\text{least squares}} k \text{ and } k'$$

$$h^{(d)}(t) = \begin{cases} 1 & \text{O-O distace of molecules } i, j \text{ is less than } R = 3.5 \text{ \AA} \\ 0 & \text{otherwise} \end{cases}$$

k and k' are the **forward** and **backward rate constants**. (Luzar, *J. Chem. Phys.* **113**, 10663 (2000); Khaliullin and Kühne, *Phys. Chem. Chem. Phys.*, **15**, 15746 (2013))



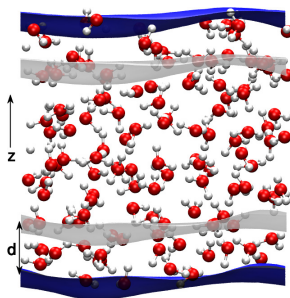
3. HB dynamics of the interface of electrolyte solutions

Definition of instantaneous interfaces

The density at $(\mathbf{r}, t)$:

$$\bar{\rho}(\mathbf{r}, t) = \sum_i \phi(|\mathbf{r} - \mathbf{r}_i(t)|; \xi), \quad \phi(\mathbf{r}; \xi) = (2\pi\xi^2)^{-3/2} \exp(-r^2/2\xi^2),$$

$\phi(\mathbf{r}; \xi)$ is a normalized **Gaussian function**. $\bar{\rho}(\mathbf{r}, t) = \rho_c$ defines the **instantaneous interface**, ρ_c is a reference density.

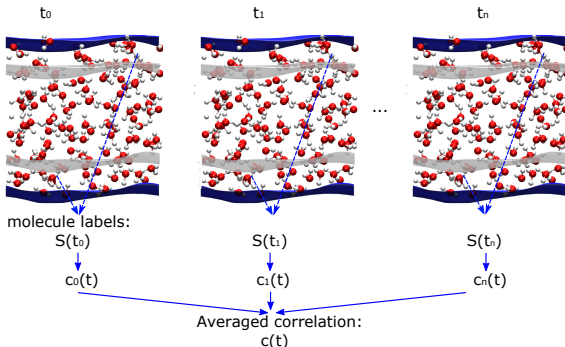


A slab of water with the instantaneous interface represented as a blue mesh on the upper and lower boundary.

3. HB dynamics of the interface of electrolyte solutions

Extracting interfacial bonds (1): Interfacial Molecule Sampling (IMS)

The **first method** to extract interfacial HB dynamics: Directly sampling. (**Slower** than the real interfacial HB dynamics.)



- (1) **Define interfacial layers** with thickness d ;
- (2) **Identify molecules** in these layers, for sampling time $t_0, \dots, t_n$;
- (3) **Calculate** functions $c_i(t)$ of those molecules;
- (4) **Average:** $c(t) = \sum_{i=1}^n c_i(t)/n$.

Extracting interfacial bonds (2): Interfacial HB population

The **second method** to extract the interfacial HB dynamics.

- ▶ Define **IHB population**

$$h^{(s)}[r(t)] = \begin{cases} 1 & i, j \text{ are H-bonded, and} \\ & i, j \text{ are inside the interfacial layer} \\ 0 & \text{otherwise} \end{cases}$$

Then, define functions

$$c^{(s)}(t) = \langle h^{(s)}(0)h^{(s)}(t) \rangle / \langle h^{(s)} \rangle$$

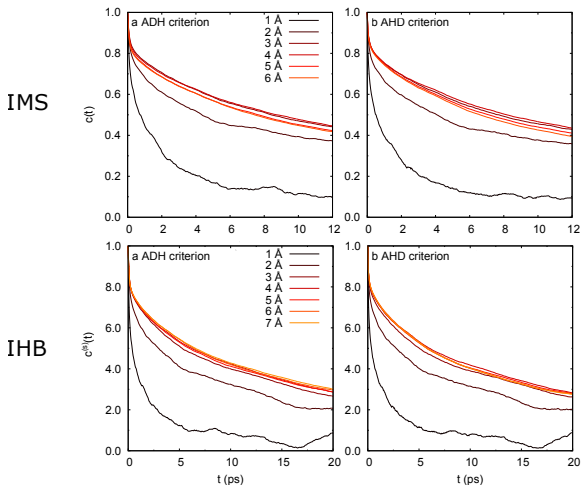
$$n^{(s)}(t) = \langle h^{(s)}(0)[1 - h^{(s)}(t)]h^{(d,s)} \rangle / \langle h^{(s)} \rangle$$

$$k^{(s)}(t) = -\frac{dc^{(s)}(t)}{dt}$$

Difference from the IMS method: IHB dynamics are **faster than the real interfacial HB dynamics**. (G. Huang, J. Huang, *arXiv:2204.13941*, <https://arxiv.org/abs/2204.13941>)

3. HB dynamics of the interface of electrolyte solutions

Both correlation functions ($c(t)$, $c^{(s)}(t)$) converge as d is larger than 4 Å.

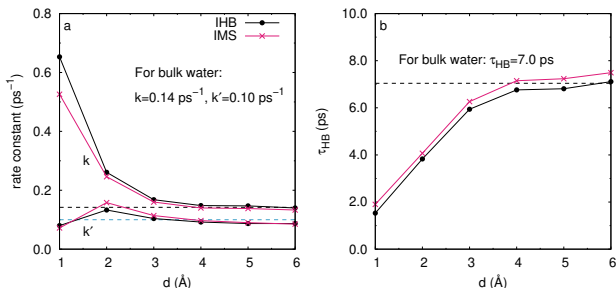


$c(t)$ ($c^{(s)}(t)$) for instantaneous interfacial H-bonds with different thicknesses.

3. HB dynamics of the interface of electrolyte solutions

Thickness of the water/vapor interface is 4 Å.

As d increases, values of k and k' converge to bulk values. → Thickness of the water/vapor interface is 4 Å.



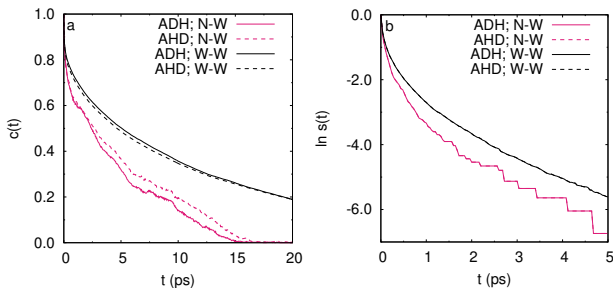
(a) Reaction rate constants k and k' and (b) the HB lifetime $\tau_{\text{HB}} = 1/k$, obtained by the IHB and the IMS method, respectively.

(G. Huang, J. Huang, *arXiv:2204.13941*, <https://arxiv.org/abs/2204.13941>)

3. HB dynamics of the interface of electrolyte solutions

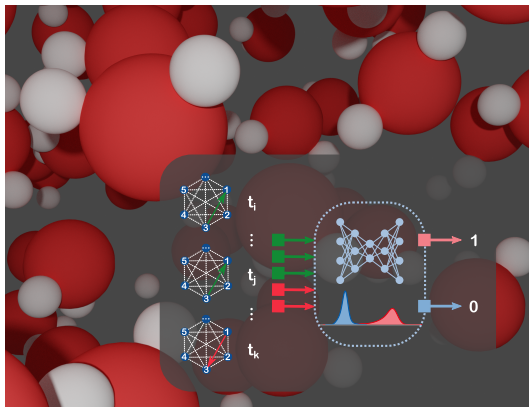
$c(t)$ and $s(t)$: NO_3^- –water bonds are weaker than water–water bonds!

$c(t)$ and $s(t)$ of **nitrate–water** (N-W) bonds decay **faster** than that of **water–water** (W-W) bonds (nitrate–water bonds are **weaker** than water–water bonds).



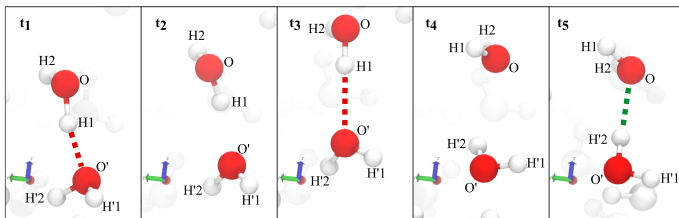
(a) $c(t)$ and (b) $\ln s(t)$ of all W-W and N-W bonds for the slab of LiNO_3 solution.

A machine learning model to classify dynamic processes in liquid water.



(J. Huang, G. Huang, S. B. Li, *ChemPhysChem*, (2022), **23**, e202100599 (Front cover))

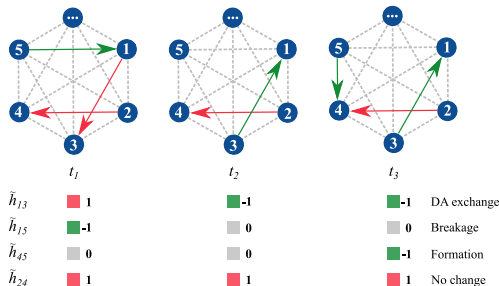
Donor-Acceptor exchange processes in water



A typical interchange process, where two water molecules exchange their roles as H-bond donor and acceptor via water molecules' reorientation in an concerted manner.

(J. Huang, G. Huang, S. B. Li, *ChemPhysChem* 2022, **23**, e202100599)

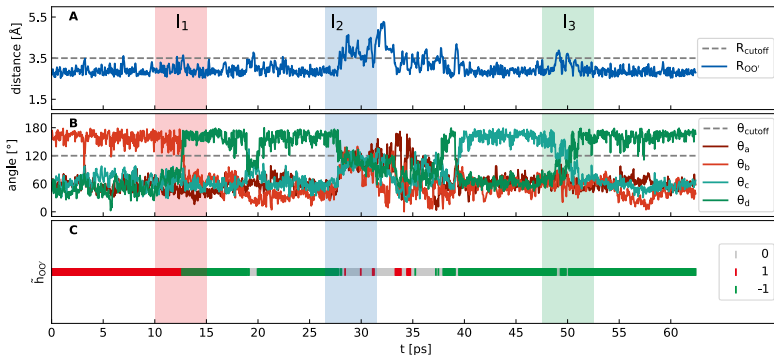
Dynamic graph representation of H-bond networks



Dynamic graph representation of the H-bond network in simulated bulk water. Nodes represent **water molecules**; solid red or green arrows represent **H-bonds**. The colors **red**, **grey**, and **green** indicate $\tilde{h}_{ij}=1$, $\tilde{h}_{ij}=0$, and $\tilde{h}_{ij}=-1$, respectively.

$$\tilde{h}_{ij}(t) = \begin{cases} 1 & \text{H-bonded, } i \text{ is the donor} \\ 0 & \text{Not H-bonded} \\ -1 & \text{H-bonded, } j \text{ is the donor} \end{cases}$$

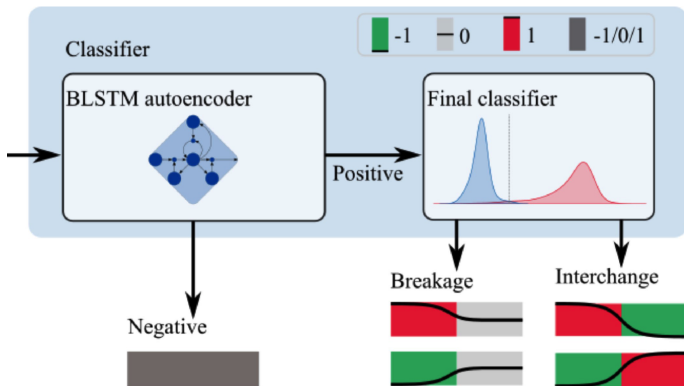
H-bond configuration change processes



DA exchange (I_1), diffusion (I_2), and HH exchange (I_3) process for one typical Q-bond in bulk water.

(J. Huang, G. Huang, S. B. Li, *ChemPhysChem* 2022, **23**, e202100599)

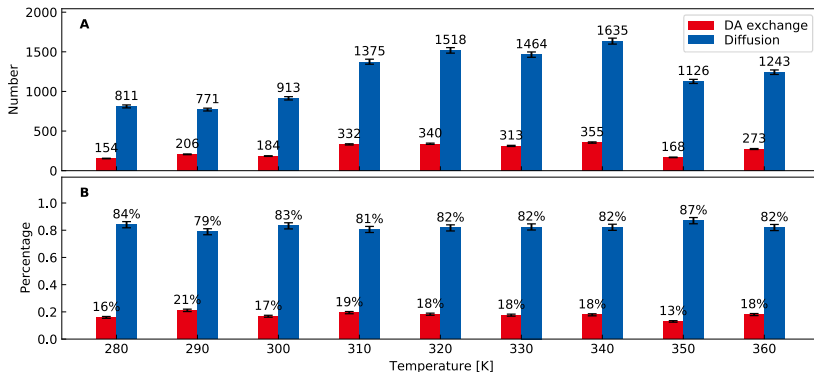
To classify the dynamical processes!



The classifier includes a **BLSTM auto encoder** to separate the positive and negative sequences and a **final classifier** to distinguish DA exchange and breakage processes.

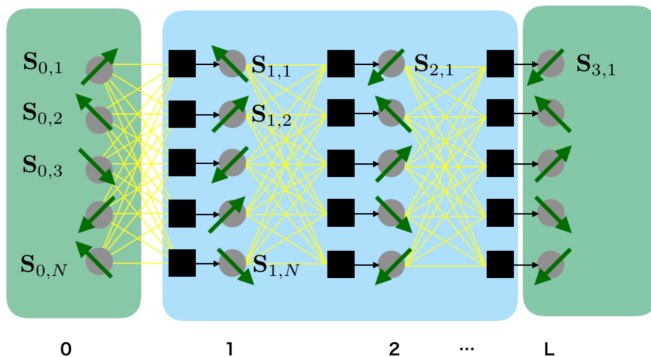
(J. Huang, G. Huang, S. B. Li, *ChemPhysChem* 2022, **23**, e202100599)

Proportions of DA exchange



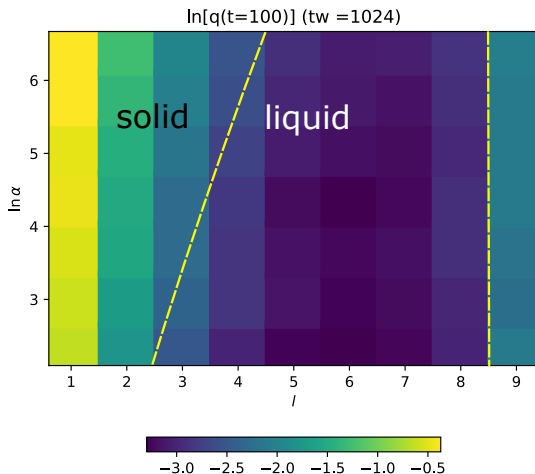
The **number** (A) and **proportion** (B) of DA exchange and diffusion processes determined by the RNN-based classifier at different temperatures. The **relative ratio** (0.25) of DA exchange to diffusion basically does not depend on temperature.

A L -layer Neural Network (NN).



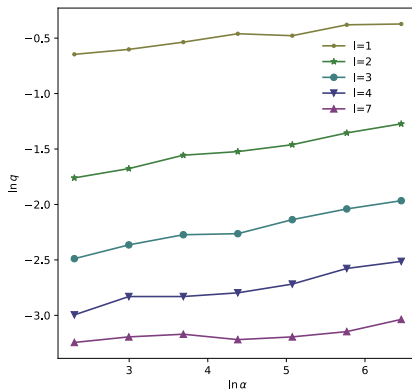
A L -layer NN. Totally $L + 1$ layers. ($L = 10$) $l = 0$ denotes input layer; $l = L$ denotes the output layer. (H. Yoshino, SciPost Phys. Core 2, 005 (2020))

$$\ln q(\ln \alpha, l)$$



Similarity between **two DNN's** nodes' states, for the training process of DNN on the **MNIST dataset**. The two DNN have the same structure and the same initial condition.

$\ln q(\ln \alpha)$



Similarity between **two** DNN's nodes' states, for the training process of DNN on the **MNIST dataset**. The two DNN have the same structure and the same initial condition.

Conclusions

- ▶ The experimental spectra (**lower intensity for bonded OH stretch**) can be explained by the calculations. The **lower intensity** is due to the presence of a **water separated ion pair** where the nitrate resides on the surfaces and the alkali cation is in the subsequent layer.
- ▶ **Thickness** of the water/vapor interface: **4 Å**.
- ▶ Based on **instantaneous interface**, combination of IHB and IMS provide a basis for determining the **thickness** of interface layer.
- ▶ The **relative ratio** of DA exchange to diffusion basically is about **0.25** and it does **not depend on temperature**.

Acknowledgment

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- ▶ Hajime Yoshino (Osaka University)
- ▶ **Yuliang Jin** (Institute of Theoretical Physics, CAS)

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THANK YOU!